PROGRAMMING

Python Interview Questions & Coding Tips

Python Fundamentals, Memory & Language Internals

1. Python Data Model & Mutability

- Mutable Types: list, dict, set, bytearray (modified in-place).
- Immutable Types: int, float, bool, str, tuple, frozenset (creates new object upon modification).
- Memory Management: Reference counting combined with generational cyclic garbage collector.
- Shallow vs. Deep Copy: 'copy.copy()' copies container pointers; 'copy.deepcopy()' recursively clones all nested objects.

2. Global Interpreter Lock (GIL) & Concurrency

- What is GIL? A mutex that prevents multiple native threads from executing Python bytecode simultaneously.
- CPU-bound tasks: Use 'multiprocessing' module to bypass the GIL by spawning independent OS processes.
- I/O-bound tasks: Use 'threading' or 'asyncio' (coroutines with event loop) for asynchronous non-blocking performance.
- 'with' Statement: Context Managers implementing '__enter__' and '__exit__' for safe resource allocation.

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Advanced Python Constructs & Best Practices

3. Decorators, Generators & Comprehensions

- Decorator: A callable that wraps another function to augment behavior without modifying source code.
- Generator: Functions yielding values lazily using 'yield' keyword; conserves memory for large data pipelines.
- List & Dict Comprehensions: [x**2 for x in range(10) if x % 2 == 0] (faster than manual loops).
- *args and **kwargs: Variable positional arguments (tuple) and keyword arguments (dictionary).

4. Object-Oriented Python & Magic Methods

- Magic (Dunder) Methods: '__init__' (initializer), '__str__' (user readable), '__repr__' (unambiguous code representation).
- Method Resolution Order (MRO): C3 Linearization algorithm for multiple inheritance; inspect via 'Class.__mro__'.
- '@property' decorator: Pythonic encapsulation to provide managed getter and setter attributes.
- Popular Libraries Overview: NumPy (vectorized arrays), Pandas (DataFrames), Scikit-Learn (ML pipelines).